INDIUM

(Data in metric tons of indium content unless otherwise noted)

Domestic Production and Use: Indium was not recovered from ores in the United States in 2017. Several companies produced indium products—including alloys, compounds, high-purity metal, and solders—from imported indium metal. Production of indium tin oxide (ITO) continued to account for most of global indium consumption. ITO thin-film coatings were primarily used for electrical conductive purposes in a variety of flat-panel displays—most commonly liquid crystal displays (LCDs). Other indium end uses included alloys and solders, compounds, electrical components and semiconductors, and research. Based on an average of recent annual import levels, estimated domestic consumption of refined indium was 120 tons in 2017. The estimated value of refined indium consumed domestically in 2017, based on the average free market price, was about $26 million.

Salient Statistics—United States:	**2013**	**2014**	**2015**	**2016**	**2017[e]**
Production, refinery	—	—	—	—	—
Imports for consumption	97	123	140	160	120
Exports	NA	NA	NA	NA	NA
Consumption, estimated[1]	97	123	140	160	120
Price, annual average, dollars per kilogram:					
New York dealer[2]	570	705	520	345	360
Free market[3]	NA	NA	410	240	205
Net import reliance[4] as a percentage of estimated consumption	100	100	100	100	100

Recycling: Indium is most commonly recovered from ITO scrap in Japan and the Republic of Korea. A significant quantity of new scrap was recycled domestically; however, data on the quantity of secondary indium recovered from scrap were not available.

Import Sources (2013–16): Canada, 23%; China, 22%; France, 11%; Republic of Korea, 11%; and other, 33%.

Tariff: Item	Number	Normal Trade Relations 12–31–17
Unwrought indium, including powders	8112.92.3000	Free.

Depletion Allowance: 14% (Domestic and foreign).

Government Stockpile: None.

Events, Trends, and Issues: The 2017 estimated average free market price of indium was $205 per kilogram. The average monthly price began the year at $210 per kilogram in January and remained level for the first 3 months of the year, after which it decreased from April to July, falling to $193 per kilogram. The average monthly price then increased through October, rising to $215 per kilogram.

Prepared by **C. Schuyler Anderson [(703) 648–4985, csanderson@usgs.gov]**

INDIUM

In China, Fanya Metal Exchange warehouses reportedly held 3,600 tons of indium as of November 2017, and no information was available as to when the inventory would be released into the market. A China-based tin producer partnered with a United States-based producer of indium-based products to construct an ITO production plant in China. ITO is a semiconducting compound used in flat-panel displays for smartphones, monitors, and other electronic devices. In early 2017, China's Ministry of Commerce implemented an export license system and eliminated the previously used export quota system, which limited the amount of indium that could be exported. This new policy was expected to encourage exports of indium.

A Belgium-based zinc producer resumed production of indium at its zinc plant in Auby, France, in early 2017. The indium production plant at the zinc smelter had been shut down in November 2015 owing to a fire. The Auby plant previously produced 89 tons of indium between 2012 and 2014.

A Belgium-based materials technology company announced in late 2017 that they would be shutting down their ITO production operations in Providence, RI, by the end of 2017, and selling off its remaining ITO production to its joint-venture partner in China, a producer of minor-metals-based products.

World Refinery Production and Reserves:

	Refinery production[e]		Reserves[5]
	2016	2017	
United States	—	—	Quantitative estimates of reserves are not available.
Belgium	20	20	
Canada	71	70	
China	300	310	
France	—	20	
Japan	70	70	
Korea, Republic of	210	215	
Peru	10	10	
Russia	5	5	
World total (rounded)	680	720	

World Resources: Indium is most commonly recovered from the zinc-sulfide ore mineral sphalerite. The indium content of zinc deposits from which it is recovered ranges from less than 1 part per million to 100 parts per million. Although the geochemical properties of indium are such that it occurs in trace amounts in other base-metal sulfides—particularly chalcopyrite and stannite—most deposits of these minerals are subeconomic for indium.

Substitutes: Antimony tin oxide coatings have been developed as an alternative to ITO coatings in LCDs and have been successfully annealed to LCD glass; carbon nanotube coatings have been developed as an alternative to ITO coatings in flexible displays, solar cells, and touch screens; PEDOT [poly(3,4-ethylene dioxythiophene)] has also been developed as a substitute for ITO in flexible displays and organic light-emitting diodes; and silver nanowires have been explored as a substitute for ITO in touch screens. Graphene has been developed to replace ITO electrodes in solar cells and also has been explored as a replacement for ITO in flexible touch screens. Researchers have developed a more adhesive zinc oxide nanopowder to replace ITO in LCDs. Gallium arsenide can substitute for indium phosphide in solar cells and in many semiconductor applications. Hafnium can replace indium in nuclear reactor control rod alloys.

[e]Estimated. NA Not available. — Zero.
[1]Assumed to equal imports.
[2]Price is based on 99.99%-minimum-purity indium; delivered duty paid U.S. buyers; in minimum lots of 50 kilograms. Source: Platts Metals Week.
[3]Price is based on 99.99%-minimum-purity indium at warehouse (Rotterdam). Source: Metal Bulletin.
[4]Defined as imports – exports.
[5]See Appendix C for resource and reserve definitions and information concerning data sources.